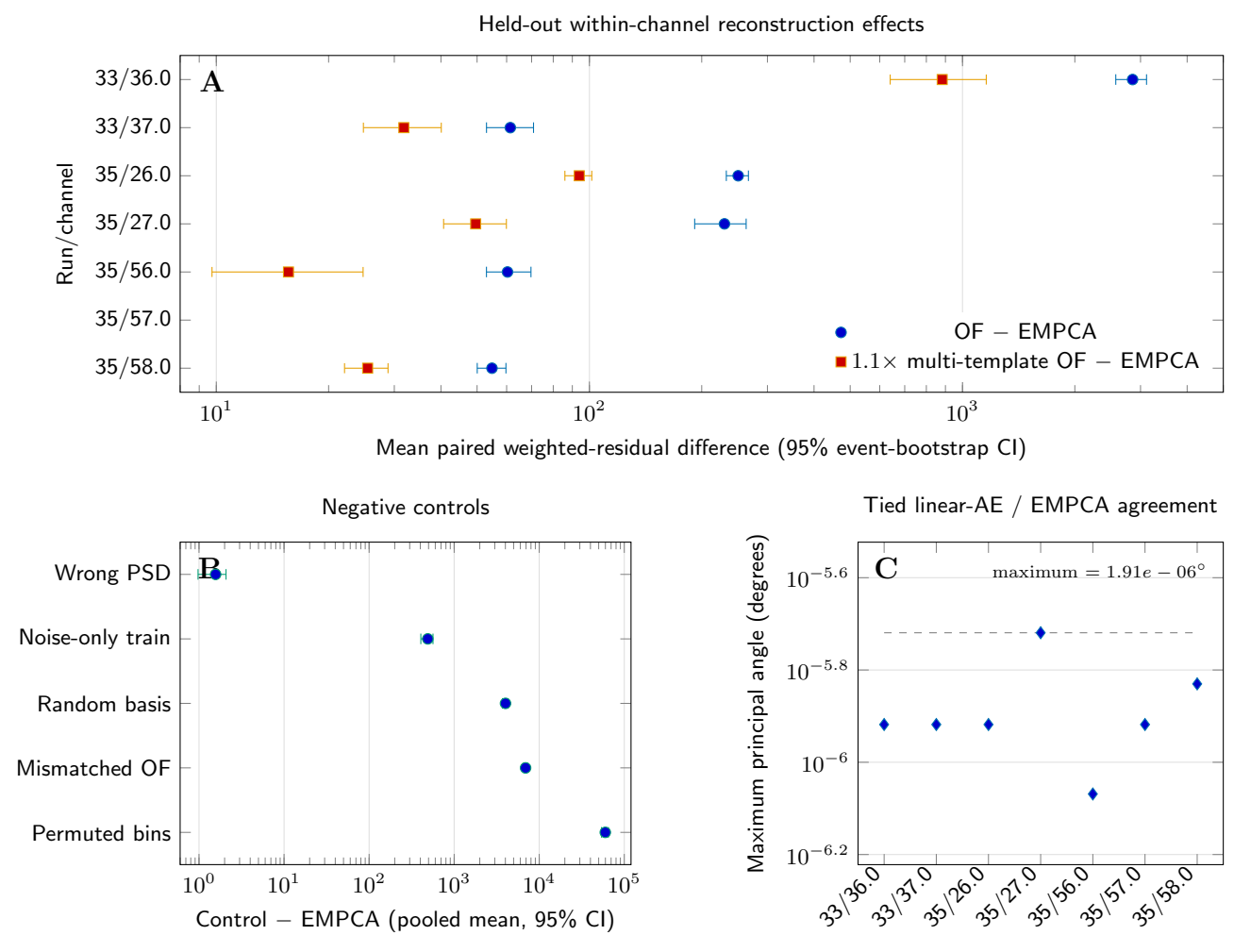

Held-out within-channel reconstruction effects
A
33/36.0
33/37.0
35/26.0
35/27.0
35/56.0
35/57.0
35/58.0
Run/channel
OF − EMPCA
1.1× multi-template OF − EMPCA
10^1
10^2
10^3
Mean paired weighted-residual difference (95% event-bootstrap CI)
Negative controls
B
Wrong PSD
Noise-only train
Random basis
Mismatched OF
Permuted bins
10^0
10^1
10^2
10^3
10^4
10^5
Control − EMPCA (pooled mean, 95% CI)
Tied linear-AE / EMPCA agreement
C
maximum = 1.91e − 06°
Maximum principal angle (degrees)
$10^{-5.6}$
$10^{-5.8}$
10^{-6}
$10^{-6.2}$
33/36.0
33/37.0
35/26.0
35/27.0
35/56.0
35/57.0
35/58.0